



Solution Guide: Gather Evidence

- The first step is to navigate into the `/03-student/day2/Gathering_Evidence/` folder on our virtual machines. To do this, run the following command:
 - `cd /03-student/day2/Gathering_Evidence/`
- In the `Gathering_Evidence` folder, make a directory called `Sugar_evidence` by running the following:
 - `mkdir Sugar_evidence`
- Next, move into the email directory:
 - `cd email`
- List the emails referencing Sugar, and place the results in a file called `Sugar_email_evidence`:
 - `grep -il sugar * > sugar_email_evidence`
- Next, add the number of emails that contain Sugar to the same file:
 - `grep -il sugar * | wc -l >> sugar_email_evidence`
- The next step is to place the following data into a new file, called `sugar_web_evidence`:
 - The web log file that contains Sugar's IP address
 - The number of Sugar IPs addresses found in this file
- Access the `web_logs` directory by going back a directory (`cd ..`), then by running the following:
 - `cd web_logs`
- To find the log files that have the IP address of `13.16.23.234` and place them in a new file called `sugar_web_evidence`, run the following:
 - `grep -il 13.16.23.234 * > sugar_web_evidence`
- Next, to append the number of records that contain the IP address to this file, run the following:
 - `grep -i 13.16.23.234 * | wc -l >> sugar_web_evidence`
- Now, move both evidence files to the `Sugar_evidence` directory that we created:
 - `cd ../../`
 - `mv ./Gathering_Evidence/email/sugar_email_evidence ./Gathering_Evidence/Sugar_evidence/`
 - `mv ./Gathering_Evidence/web_logs/sugar_web_evidence ./Gathering_Evidence/Sugar_evidence/`
- Next, navigate to the `Sugar_evidence` directory, and concatenate the files into a single file called `Sugar_evidence_for_authorities`:
 - `cd Gathering_Evidence/Sugar_evidence/`

- `cat sugar_email_evidence sugar_web_evidence > Sugar_evidence_for_authorities`
- The last step is to confirm that the file has the correct data for the authorities. Run the following command:
 - `cat Sugar_evidence_for_authorities`
- The results should be as follows:

Unset

```
email5
email7
email8
3
0518weblog
22
```

- While this file contains all the data the authorities need, this is not obvious. In the next lesson, we will learn how to edit the contents of a file to further describe the evidence.